WEEK 9 – ACTIVITY

Title: Types of Networks & Their Use Cases

Introduction: In today's digital world, computer networks play a very important role in communication and data sharing. Different types of networks are used based on the size, coverage area, and purpose. Understanding these network types helps in designing efficient and secure communication systems.

In this activity, I explored three major types of networks: LAN (Local Area Network), WAN (Wide Area Network), and PAN (Personal Area Network). I compared their features and analyzed their real-world applications.

Types of Networks:

- **Local Area Network (LAN)**
  A Local Area Network (LAN) is a network that connects devices within a **small geographical area** such as a home, office, or college campus.
  - It provides high-speed communication
  - Used for sharing files, printers, and internet
  - Typically managed by a single organization

  Example: Computer labs in colleges, office networks

- **Wide Area Network (WAN)**
  A Wide Area Network (WAN) covers a **large geographical area** such as cities, countries, or even the entire world.
  - Connects multiple LANs together
  - Provides long-distance communication
  - Slower compared to LAN due to large coverage

  Example: The Internet

- **Personal Area Network (PAN)**
  A Personal Area Network (PAN) is a small network **used for personal devices.**
  - Covers very short distances (few meters)
  - Used for connecting personal gadgets
  - Uses technologies like Bluetooth or USB

Example: Mobile connected to smartwatch via Bluetooth

Comparison Table:

Network Type	Full Form	Coverage Area	Speed	Ownership	Example
LAN	Local Area Network	Small (building)	High	Private	College Lab
WAN	Wide Area Network	Large (global)	Medium	Public/Private	Internet
PAN	Personal Area Network	Very small	Low	Personal	Bluetooth

Real-World Use Cases

- **LAN Use Case:** LAN is commonly used in offices and educational institutions where multiple computers need to be connected within a building. It allows fast file sharing, printer access, and centralized control.
- **WAN Use Case:** WAN is used by large organizations such as banks and multinational companies to connect their branches across different cities and countries.
- **PAN Use Case:** PAN is used in everyday life for connecting personal devices such as wireless earphones, fitness bands, and mobile phones.

Advantages and Limitations

- LAN : High speed, Secure, Limited coverage
- WAN: Global connectivity, Enables communication worldwide, Slower and costly
- PAN: Easy to use, Low power consumption, Very limited range

Learning Outcome

Through this activity, I understood how different types of networks are designed based on requirements. I learned how LAN, WAN, and PAN differ in terms of

coverage, speed, and application. This knowledge is essential for understanding networking fundamentals and cyber security concepts.

Conclusion

In conclusion, each type of network serves a specific purpose. LAN is suitable for small areas, WAN is used for large-scale communication, and PAN is ideal for personal device connectivity. Understanding these networks helps in designing efficient and secure systems in real-world scenarios.